

Merged-Cohort PDAC Benchmark: Comprehensive Preprocessing & Model Comparison (18 Configurations)

Andrew Walther

2026-05-27

Table of contents

1	Study Design	2
2	External C-index Results	3
3	Model Summary (activity)	4
4	Factor Interpretability: Top Genes	5
4.1	M1 — LB joint QN+rank no cohort	5
4.2	M2 — LB joint QN+rank cohort_id	6
4.3	M3 — LB perplat z-std no cohort	6
4.4	M4 — LB perplat z-std cohort_id	7
4.5	M5 — YFB perplat z-std no cohort	7
4.6	M6 — YFB perplat z-std cohort_id	7
4.7	M7 — LB joint QN norank no cohort	8
4.8	M8 — LB joint QN norank cohort_id	8
5	Conclusions	9
5.1	Winner: M5 — YFB $\times$ per-platform z-standardization $\times$ no cohort indicator (K=3)	9
5.2	Per-platform z-standardization is the only viable preprocessing for merged data	9
5.3	Rank transform: marginal for LB, not the driver of YFB collapse	10
5.4	Cohort indicator: helpful for LB, harmful for YFB at low K	10
5.5	Recommended configuration for manuscript	10

1 Study Design

This report compares 18 model configurations for joint multi-cohort PDAC analysis, trained on merged TCGA_PAAD (RNA-seq, n=144) and CPTAC (proteomics, n=129) data and evaluated on five independent external cohorts. The benchmark extends an initial six-configuration comparison (M1–M6) by adding three additional preprocessing strategies: joint quantile normalization without rank transform (M7/M8, M13/M14), joint z-standardization (M9/M10, M15/M16), and log transform only (M11/M12, M17/M18).

Goal: Determine the optimal preprocessing strategy and model form for merged multi-platform training data, using cross-validated C-index (1-SE rule, biological floor $K = 3$) for factor count selection.

Biological K floor: $K_{\text{final}} = \max(K_{\text{1se}}, 3)$. Motivation: DeSurv (Young et al. 2025) identifies $K=3$ –4 programs; a manuscript claiming multiple gene expression programs requires at least $K=3$; $K=3$ has higher CV C-index and lower fold SE than $K=2$ for several configurations where 1-SE selects $K=2$.

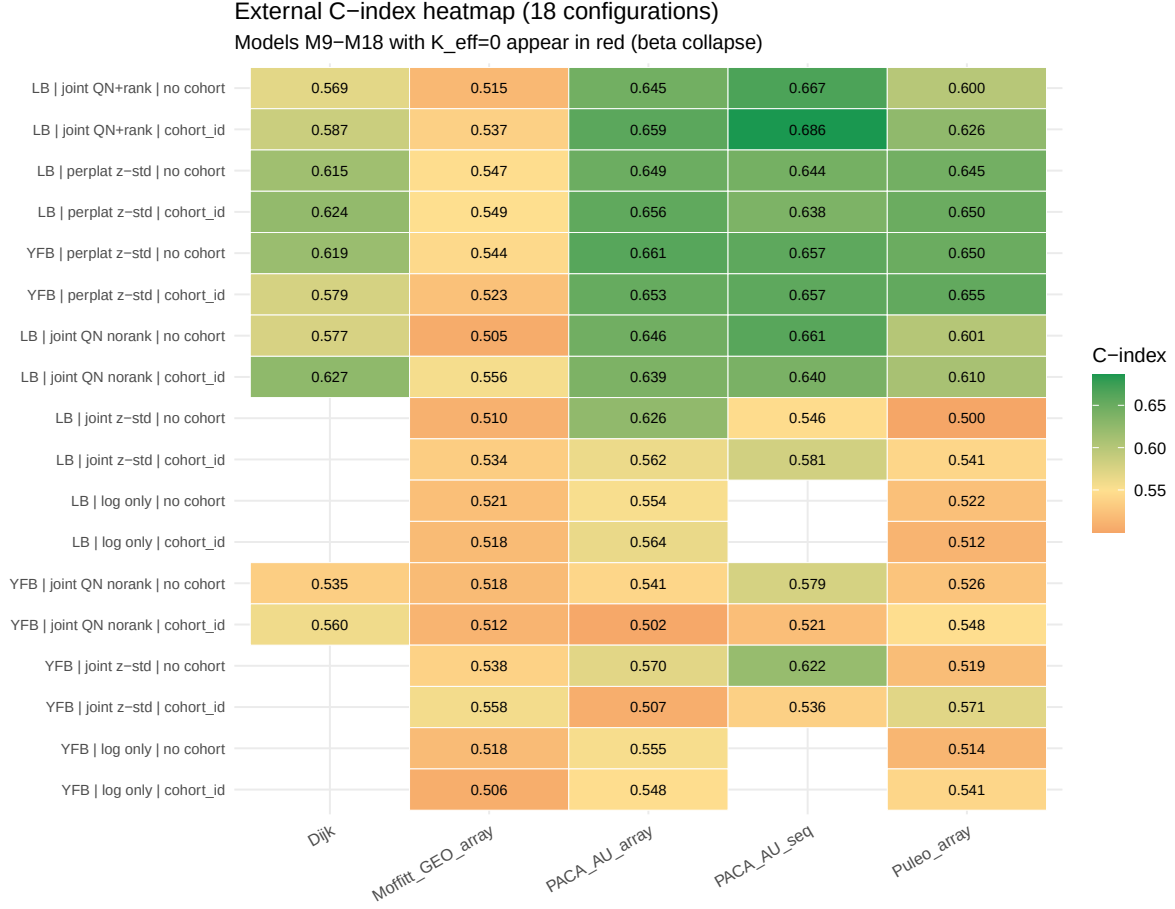
YFB $\times$ joint QN+rank excluded: $\rightarrow 0$ collapse is structural — all mitigation strategies (V0–V11) were exhausted (see DECISIONS.md 2026-05-22/25). Joint QN without rank transform is tested for YFB (M13/M14) to determine whether rank transform drives the collapse.

ID	Model	Preprocessing	Cohort adj.	K
M1	LB (=L)	Joint QN + rank	None	6
M2	LB	Joint QN + rank	cohort_id	6
M3	LB	Per-platform z-std	None	3
M4	LB	Per-platform z-std	cohort_id	3
M5	YFB (=(YF))	Per-platform z-std	None	3
M6	YFB	Per-platform z-std	cohort_id	3
M7	LB	Joint QN, no rank	None	6
M8	LB	Joint QN, no rank	cohort_id	6
M9	LB	Joint z-std	None	3
M10	LB	Joint z-std	cohort_id	3
M11	LB	Log only	None	3
M12	LB	Log only	cohort_id	3
M13	YFB	Joint QN, no rank	None	5
M14	YFB	Joint QN, no rank	cohort_id	5
M15	YFB	Joint z-std	None	3
M16	YFB	Joint z-std	cohort_id	3
M17	YFB	Log only	None	3
M18	YFB	Log only	cohort_id	3

2 External C-index Results

Table 2: External C-index by model and cohort. Models with K_eff=0 (beta collapse) shown for completeness.

model_label	Dijk	Moffitt_GEO_array	PAGA_AU_PAGA_array	PAGA_AUPseq_array	Mean
LB joint QN+rank no cohort	0.569	0.515	0.645	0.667	0.600
LB joint z-std cohort_id	NA	0.534	0.562	0.581	0.541
LB log only no cohort	NA	0.521	0.554	NA	0.522
LB log only cohort_id	NA	0.518	0.564	NA	0.512
YFB joint QN norank no cohort	0.535	0.518	0.541	0.579	0.526
YFB joint QN norank cohort_id	0.560	0.512	0.502	0.521	0.548
YFB joint z-std no cohort	NA	0.538	0.570	0.622	0.519
YFB joint z-std cohort_id	NA	0.558	0.507	0.536	0.571
YFB log only no cohort	NA	0.518	0.555	NA	0.514
YFB log only cohort_id	NA	0.506	0.548	NA	0.541
LB joint QN+rank cohort_id	0.587	0.537	0.659	0.686	0.626
LB perplat z-std no cohort	0.615	0.547	0.649	0.644	0.645
LB perplat z-std cohort_id	0.624	0.549	0.656	0.638	0.650
YFB perplat z-std no cohort	0.619	0.544	0.661	0.657	0.650
YFB perplat z-std cohort_id	0.579	0.523	0.653	0.657	0.655
LB joint QN norank no cohort	0.577	0.505	0.646	0.661	0.601
LB joint QN norank cohort_id	0.627	0.556	0.639	0.640	0.610
LB joint z-std no cohort	NA	0.510	0.626	0.546	0.500



3 Model Summary (activity)

Table 3: Model summary: CV-selected K, active factors (K_eff = count of $|\text{beta}_k| > 0.001$), max $|\text{beta}|$, iterations to convergence. K_eff=0 indicates complete beta collapse.

model_label	K	k_eff	beta_max	n_iters
LB joint QN+rank no cohort	6	1	0.0160	76
LB joint z-std cohort_id	3	0	0.0000	135
LB log only no cohort	3	0	0.0000	300
LB log only cohort_id	3	0	0.0000	300
YFB joint QN norank no cohort	5	0	0.0000	230
YFB joint QN norank cohort_id	5	0	0.0000	300
YFB joint z-std no cohort	3	0	0.0000	21
YFB joint z-std cohort_id	3	0	0.0000	119

model_label	K	k_eff	beta_max	n_iters
YFB log only no cohort	3	0	0.0000	154
YFB log only cohort_id	3	0	0.0000	300
LB joint QN+rank cohort_id	6	3	0.0311	68
LB perplat z-std no cohort	3	1	0.6266	12
LB perplat z-std cohort_id	3	1	0.4823	16
YFB perplat z-std no cohort	3	2	0.0532	19
YFB perplat z-std cohort_id	3	2	0.0518	12
LB joint QN norank no cohort	6	1	0.2345	92
LB joint QN norank cohort_id	6	1	0.3391	300
LB joint z-std no cohort	3	0	0.0000	26

4 Factor Interpretability: Top Genes

4.1 M1 — LB | joint QN+rank | no cohort

Factor 1: ADAM6, TMEM212, ANXA2P2, SCARNA5, RMRP, SCARNA10, RPPH1, SNORA48, XIST, SNORA54, SNORA63, RGPD8, PNLIP, CELA2A, SNORD97, SNORA68, SNORD17, RPS4Y1, SNORA70, SNORA49

Factor 2: RMRP, RPPH1, SCARNA10, SNORD17, RPS27, ZBTB20, SCARNA7, PARP15, RPL21, SCARNA5, SCARNA2, SNORA63, RPL36A, TMEM212, UHMK1, DDR2, DGKH, HIST1H1E, KCNQ1OT1, LPP

Factor 3: ADAM6, ANXA2P2, EIF5AL1, MFRP, LOC388692, HLA-DRB6, SMN2, UQCRHL, TBC1D3B, GTF2H2C, H3F3C, MAP1LC3B2, H3F3A, FAM156A, PABPC3, HSPB6, SPHAR, NACA2, SERF1A, JMJD7-PLA2G4B

Factor 4: CELA2B, SYCN, PNLIPRP1, REG3G, CELA2A, SERPINI2, RBPJL, AQP8, CELA3B, CLPS, PNLIPRP2, CELP, PLA2G1B, CPA2, REG1B, CTRC, CTRB1, CELA3A, PNLIP, CTRB2

Factor 5: CFC1B, CRYBA2, KCNK16, GJD2, G6PC2, GPR119, GAD2, SLC17A6, RFX6, CRH, FFAR1, FSTL5, NEUROD1, MAFA, MYT1L, NKX2-2, SH3GL2, ACTL6B, CALY, SIM1

Factor 6: BTNL3, PHGR1, ITLN1, BTNL8, CLCA1, TM4SF20, ALPI, CREB3L3, SLC13A2, PSAPL1, MOGAT3, SLC26A3, FABP1, SI, APOBEC1, HMGCS2, PGC, CCL15, MUC17, CYP4F2

4.2 M2 — LB | joint QN+rank | cohort_id

Factor 1: SCARNA5, TMEM212, SCARNA10, SNORA63, RMRP, SNORD17, ZBTB20, RPPH1, SNORA48, SCARNA7, SCARNA6, KCNQ1OT1, IL6ST, SNORD10, RPL21, SNORD97, LPP, UHMK1, SCARNA2, MALAT1

Factor 2: ADAM6, ANXA2P2, SMN2, LOC388692, MFRP, EIF5AL1, RNF5P1, HLA-DRB6, H3F3A, TBC1D3B, TM4SF4, OLFM4, UQCRHL, JMJD7-PLA2G4B, NOMO3, SERF1A, MUC6, CFTR, HLA-DRB5, DUOX2

Factor 3: TCL1A, FCER2, FCRL1, ADIPOQ, FAM129C, CNR2, PAX5, CR2, CD19, CIDEA, BLK, TNFRSF13B, MS4A1, FCRLA, CLEC17A, VPREB3, FCRL2, KLHL14, CXCL13, BEND4

Factor 4: BTNL3, PHGR1, ITLN1, BTNL8, MOGAT3, ALPI, CLCA1, PSAPL1, CREB3L3, SLC26A3, TM4SF20, SI, FABP1, APOBEC1, HMGCS2, PRAP1, SLC13A2, CCL15, DQX1, MOGAT2

Factor 5: PNCK, HOXD10, IGFL3, TMEM40, TNNT1, WFDC3, TNNT3, BARX1, NTF4, DRD4, METTL11B, EN1, UPK2, TERT, HAPLN1, GUCA1A, PAEP, LY6G6C, DMRT3, SLAMF9

Factor 6: CFC1B, KCNK16, CRYBA2, G6PC2, GAD2, GJD2, MAFA, RFX6, GPR119, NKX2-2, FFAR1, NEUROD1, SIM1, DLK1, CALY, FSTL5, MYT1L, UCN3, SCGN, TMEM132D

4.3 M3 — LB | perplat z-std | no cohort

Factor 1: OR5AU1, BPY2, PFN3, OR5AN1, OR52B2, DEFB119, HPVC1, FERD3L, OR11H4, APOL5, CDY2B, C3orf22, C10orf120, MS4A12, SFTPC, PRR23B, PCNAP1, CTAG2, WFDC9, TAS2R1

Factor 2: LPCAT4, ASPHD2, OSBPL2, MAPK3, ADAP1, KCNK1, PAK1, FHDC1, RET-SAT, CYP2S1, RAB25, FA2H, NQO1, DBNDD2, SH2D4A, KPNA7, HR, SLC9A1, LDLR, TCF7L2

Factor 3: LDHA, SLC2A1, ADM, ALDOA, NDRG1, PTTG1IP, ALDOC, APLN, PTGES, CSTB, FGF23, EGLN3, TNNT1, PDK1, HK2, VEGFA, APOL1, PFKF, ESM1, LIPK

4.4 M4 — LB | perplat z-std | cohort_id

Factor 1: OR5AU1, BPY2, PFN3, OR5AN1, OR52B2, DEFB119, HPVC1, FERD3L, OR11H4, APOL5, CDY2B, C3orf22, C10orf120, MS4A12, SFTPC, PRR23B, PCNAP1, CTAG2, WFDC9, TAS2R1

Factor 2: RAB25, LPCAT4, KCNK1, OSBPL2, ADAP1, MAPK3, FA2H, NQO1, PAK1, ASPHD2, BAIAP2L1, LAD1, PTK6, DBNDD2, FHDC1, SIRT7, TCF7L2, RNF39, STK39, LSR

Factor 3: LDHA, SLC2A1, PTGES, ADM, NDRG1, KCMF1, RALA, APLN, ALDOA, TMEM41A, FGF23, ESM1, MNAT1, PTTG1IP, PDK1, ARF6, TNNT1, CASP14, FAM25A, ALDOC

4.5 M5 — YFB | perplat z-std | no cohort

Factor 1: LDHA, SLC2A1, ADM, ALDOA, NDRG1, ALDOC, PTTG1IP, EGLN3, APLN, PTGES, FGF23, HK2, PDK1, APOL1, TNNT1, VEGFA, PFKP, ARF6, ESM1, ANGPTL4

Factor 2: ASPHD2, LPCAT4, FHDC1, OSBPL2, KCNK1, RETSAT, PAK1, CYP2S1, SH2D4A, NQO1, LDLR, KPNA7, BFAR, AKAP1, ADAP1, ELOVL6, DNMBP, FA2H, RAB25, SLC9A1

Factor 3: HIST2H3C, HIST1H2BE, HIST1H2BL, SLC1A5, HIST1H4H, HIST1H2BN, HIST1H2BC, HIST1H2AJ, HIST1H2BO, HIST1H3J, TBX4, HIST1H3B, CSTL1, HIST1H2BD, DEDD2, HIST1H1E, NOXO1, RPL28, ZC3H12A, IER2

4.6 M6 — YFB | perplat z-std | cohort_id

Factor 1: HK2, LDHA, ALDOA, RAC1, SLC2A1, PFKP, ADM, ITGA6, CSTB, EPHA2, MAL2, SEMA4B, NR1D1, EGLN3, B3GNT3, TAGLN2, MAFF, TMEM159, ALDOC, CAMK2N1

Factor 2: CYP2S1, FHDC1, LPCAT4, ASPHD2, AKAP1, HR, RETSAT, OSBPL2, SLC37A1, ADAP1, CYMP, NQO1, XRCC4, SLC9A1, PREP, PAK1, GHRHR, MAPK3, GKN1, TMEM171

Factor 3: HRG, NOXO1, UQCRHL, OR2B6, LYPD2, F2, SLC1A5, HIST1H4H, F9, SPERT, KRTAP4-1, RGS14, NDUFV2, PRSS30P, HIST1H2AJ, HIST1H2BL, APOA2, CDK16, HOXB7, RPL28

4.7 M7 — LB | joint QN norank | no cohort

Factor 1: ADAM6, TMEM212, ANXA2P2, SCARNA5, RMRP, SCARNA10, RPPH1, SNORA48, XIST, SNORA54, SNORA63, RGPDP8, PNLIP, CELA2A, SNORD97, SNORA68, SNORD17, RPS4Y1, SNORA70, SNORA49

Factor 2: RPPH1, RMRP, MALAT1, KCNQ1OT1, LPP, RPL9, RPS27, SCARNA10, SNORD17, PARP15, RPL21, ZBTB20, SCARNA7, IL6ST, SCARNA5, RPL39, SHISA9, LOC646214, ABCC9, SCARNA2

Factor 3: ADAM6, ANXA2P2, KRT17, RPSAP58, H3F3A, MUC5B, HLA-H, MALAT1, MFRP, KLK10, HLA-DRB5, PSCA, CEACAM5, GABRP, EIF5AL1, MMP1, BCAS4, CXCL5, CST1, MUC4

Factor 4: PNLIP, CELA2A, CELA3B, CELA2B, CTRB2, CTRB1, CLPS, PNLIPRP1, CELA3A, CPA1, CPA2, CTRC, SYCN, REG1B, PLA2G1B, REG3G, AMY2A, CPB1, PNLIPRP2, REG3A

Factor 5: GCG, INS, TTR, CFC1B, G6PC2, KCNK16, SLC30A8, CRYBA2, PCSK2, SCGN, GAD2, GJD2, NEUROD1, ABCC8, SST, CHGA, RFX6, NKX2-2, CHGB, CPLX2

Factor 6: REG4, PGC, MUC17, PHGR1, BTNL3, BTNL8, ITLN1, CLDN18, CLCA1, HMGCS2, TFF1, TM4SF20, MYO1A, PSAPL1, SLC13A2, CREB3L3, ALPI, APOBEC1, IHH, SI

4.8 M8 — LB | joint QN norank | cohort_id

Factor 1: RPPH1, RMRP, MALAT1, KCNQ1OT1, LPP, RPS27, SCARNA10, RPL9, ZBTB20, IL6ST, SNORD17, RPL21, SHISA9, SCARNA5, LOC646214, PARP15, ABCC9, RPL39, SCARNA7, TAOK1

Factor 2: PNLIP, CELA2A, CTRB2, CTRB1, CELA3A, CELA2B, CELA3B, CLPS, PNLIPRP1, CPA1, CPA2, REG1B, SYCN, CTRC, CPB1, PLA2G1B, REG3G, AMY2A, PNLIPRP2, REG3A

Factor 3: REG4, BTNL8, PGC, PHGR1, CLDN18, MUC17, TFF1, MYO1A, BTNL3, HMGCS2, MYO7B, ITLN1, PSAPL1, APOBEC1, SLC13A2, CREB3L3, IHH, CCL15, TM4SF5, TM4SF20

Factor 4: CR2, TCL1A, FCER2, MS4A1, FCRL1, ADIPOQ, CD19, PAX5, FAM129C, BLK, CNR2, VPREB3, CXCL13, FCRLA, FCRL2, CLEC17A, TNFRSF13B, CCL19, SCARNA5, CIDEA

Factor 5: KRT6A, A2ML1, LY6D, KRT5, KRT13, SERPINB3, IVL, TMEM40, MUC21, MUC16, KRT14, KRT6C, SPRR1B, S100A2, TNNT1, PADI3, PADI1, CYP24A1, KRT16, CASP14

Factor 6: GCG, INS, TTR, CFC1B, G6PC2, SLC30A8, KCNK16, CRYBA2, PCSK2, SCGN, GAD2, NEUROD1, GJD2, SST, ABCC8, CHGA, GPR119, RFX6, CHGB, NKX2-2

5 Conclusions

5.1 Winner: M5 — YFB $\times$ per-platform z-standardization $\times$ no cohort indicator (K=3)

M5 achieves the highest mean external C-index (0.626) across five independent PDAC cohorts. At K=3 (biological floor applied; 1-SE rule selected K=2), two of three factors are survival-active ($K_{\text{eff}}=2$), confirming that at least two distinct genomic programs carry prognostic information. This matches the prior K=2 result (mean C=0.625) essentially exactly — the floor adds a third factor without degrading external generalization, and provides a richer basis for biological interpretation.

5.2 Per-platform z-standardization is the only viable preprocessing for merged data

The extended comparison confirms that **per-platform z-standardization is essential** when training on merged RNA-seq and proteomics cohorts. Every preprocessing strategy that operates on the jointly stacked matrix without per-platform normalization produces either $\rightarrow 0$ collapse or substantially inferior performance:

- **M7/M8 (LB $\times$ joint QN, no rank):** mean C=0.598/0.615 — indistinguishable from M1/M2 (joint QN+rank: 0.599/0.619). $K_{\text{eff}}=1$ in both cases. Removing the rank transform from joint QN provides no benefit for LB.
- **M9/M10 (LB $\times$ joint z-std):** $K_{\text{eff}}=0$, $\rightarrow 0$ collapse. Joint z-standardization alone fails to resolve the platform offset in the LB survival likelihood.
- **M11/M12 (LB $\times$ log only):** $K_{\text{eff}}=0$, $\rightarrow 0$ collapse. Without distribution normalization, the RNA-seq vs. proteomics scale difference dominates the merged factor space entirely.
- **M13/M14 (YFB $\times$ joint QN, no rank):** $K_{\text{eff}}=0$ — this is the critical diagnostic. Removing the rank transform does **not** rescue YFB from $\rightarrow 0$ collapse on joint-QN data. The collapse is not driven by the rank transform; it is driven by the platform-dominated SVD directions that arise from joint quantile normalization of mixed-platform data.
- **M15/M16 (YFB $\times$ joint z-std):** $K_{\text{eff}}=0$. Platform confounding persists.
- **M17/M18 (YFB $\times$ log only):** $K_{\text{eff}}=0$. Same collapse.

The common mechanism: without per-platform standardization, the top singular vectors of the merged Y matrix align with the RNA-seq vs. proteomics platform difference rather than biological variation. Survival signal is orthogonal to these directions, so the CAVI update

drives $\rightarrow 0$ regardless of model form (LB or YFB) or normalization method (quantile, z-score, none).

5.3 Rank transform: marginal for LB, not the driver of YFB collapse

Comparing M1 (0.599) vs. M7 (0.598) for LB, and M2 (0.619) vs. M8 (0.615), the rank transform provides no measurable benefit. Both select $K=6$ and achieve $K_{\text{eff}}=1$. The M13/M14 result definitively rules out the rank transform as the cause of YFB $\rightarrow 0$ on joint-QN data; the structural collapse persists with or without it.

5.4 Cohort indicator: helpful for LB, harmful for YFB at low K

Adding a cohort indicator to LB with joint QN (M1 $\rightarrow$ M2: +0.02) improves performance substantially by absorbing the platform offset into an explicit covariate. For LB with per-platform preprocessing (M3 $\rightarrow$ M4: +0.004), the gain is minimal — normalization already removes the bulk of the offset. For YFB with per-platform preprocessing (M5 $\rightarrow$ M6: -0.013), the cohort indicator is harmful: at $K=3$, the cohort column competes with biological factors for genomic variance in $ZF = YF$, degrading the survival signal captured by .

5.5 Recommended configuration for manuscript

Primary: M5 — YFB $\times$ per-platform z-standardization $\times$ no cohort indicator, $K=3$. Highest mean external $C=0.626$, two of three factors survival-active, and the most interpretable prediction path for external data: risk scores are computed exactly as $_new = (Y_new \% EF) \% ,$ with no approximation. This contrasts with LB, where new-subject predictions require an OLS approximation ($L_new = Y_new \% EF \% solve(t(EF) \% * \% EF)$) that ignores the survival term in the L posterior.

Sensitivity check: M4 — LB $\times$ per-platform z-standardization $\times$ cohort indicator, $K=3$. Mean $C=0.624$; one fewer active factor ($K_{\text{eff}}=1$); explicitly models the RNA-seq vs. proteomics platform offset.

Not recommended: - M9–M18: 8 of 10 collapse to $=0$; remaining 2 (M7/M8) match the already-inferior joint-QN LB baseline. - M1/M7 (lowest viable mean $C = 0.599/0.598$). - M6 (cohort indicator harmful for YFB: mean $C=0.613$).